

Section 4B. Bayes Classifier and Local Averaging

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Classification Problem: Bayes Classifier

- ▶ **Bayes optimal classifier:**

- ▶ Assuming that we knew the conditional class probabilities, the *Bayes Optimal Classifier* (BOC) $C(\mathbf{x})$ is given by

$$C(\mathbf{x}) = \arg \max_k p_k(\mathbf{x})$$

In other words, given an input $\mathbf{x}$, the BOC assigns the label k that maximizes the conditional class probability. We will show later on that the BOC minimizes the classification error rate...

- ▶ *Example:* In the previous slide, since $p_1(x) \geq p_0(x)$ for $x \geq 0$, the Bayes optimal classifier is

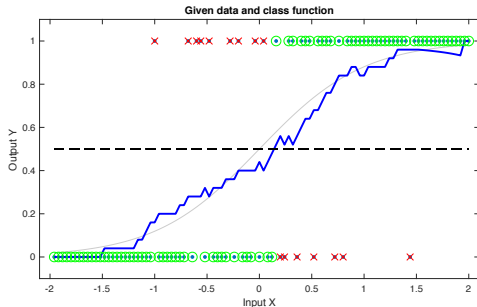
$$C(x) = \begin{cases} 1 & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

- ▶ In practice, we do not know the conditional class probabilities; in this situation, how can we find the Bayes optimal classifier?...

Classification Problem: Local Averaging

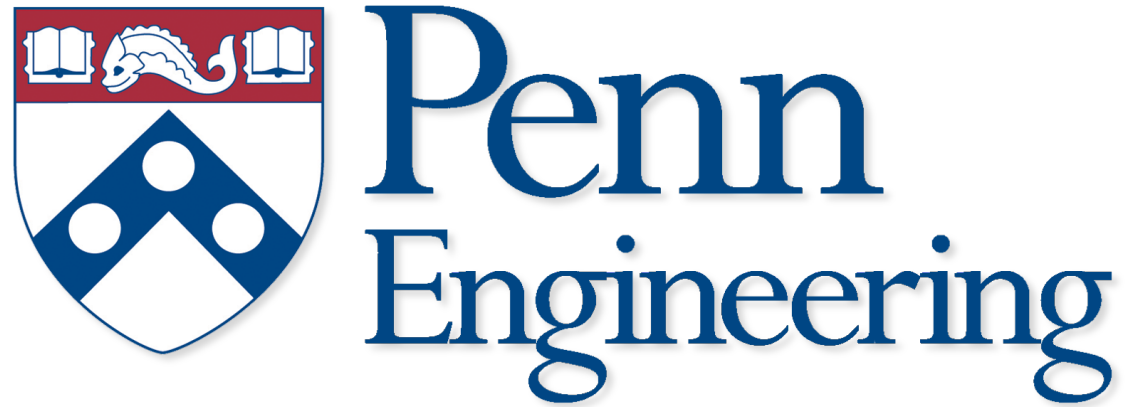
► Numerical results:

- We can numerically estimate the conditional class probability $\hat{p}_1(x)$ from data using local averaging (see blue curve in figure below)
- Using the estimate $\hat{p}_1(x)$, we can find an approximation of the Bayes optimal classifier $\hat{C}(x) = 1$ for $\hat{p}_1(x) \geq \hat{p}_0(x) \iff \hat{p}_1(x) \geq 1/2$; $\hat{C}(x) = 0$ otherwise



Curse of Dimensionality

- ▶ ***Curse of dimensionality***: However, when $\mathbf{x}$ is a high-dimensional vector, local averaging does not provide a reliable estimate of the conditional class probabilities. To overcome this issue, we will use structured models to represent $p_k(\mathbf{x})$...



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